

Lecture 3 : Inheritance and Polymorphism

1. Inheritance

1.1 What is Inheritance?

- Real-world objects are often related in parent-child relationships.
- Example: Object A (Parent) and Object B (Child) share properties.
- In programming, this relationship is mimicked using **Inheritance**.

1.2 Real-Life Example: Car Hierarchy

- **Parent Class:** Car (Generic)
 - Common attributes:
 - Brand
 - Model
 - IsEngineOn
 - CurrentSpeed
 - Common behaviors:
 - startEngine()
 - stopEngine()
 - accelerate()
 - brake()
- **Child Classes:**
 - **ManualCar** (inherits Car)
 - Specific attribute: CurrentGear

- Specific behavior: shiftGear()
- **ElectricCar** (inherits Car)
 - Specific attribute: BatteryPercentage
 - Specific behavior: chargeBattery()

1.3 C++ Syntax

```
class ManualCar : public Car { ... };  
class ElectricCar : public Car { ... };
```

- **public** inheritance maintains access specifiers.
- **private** and **protected** alter accessibility.

1.4 Access Specifiers in Inheritance

- **public:**
 - Public members stay public.
 - Protected members stay protected.
- **protected:**
 - Public and protected members become protected.
- **private:**
 - All inherited members become private.
- **Private members** of parent class are **never inherited**.

// See code section for full code example.

2. Polymorphism

2.1 What is Polymorphism?

- Derived from: **"Poly" (many) + "Morph" (forms) = many forms.**
- **One stimulus → different responses** based on object/situation.

2.2 Two Real-Life Scenarios:

- **Scenario 1:**
 - Different animals (Duck, Human, Tiger) all have a `run()` behavior.
 - Each performs it differently.
- **Scenario 2:**
 - Same human `run()`s differently based on context (tired vs chased).

2.3 Types of Polymorphism in Programming:

- **Static Polymorphism – Compile-time**
 - Achieved via **Method Overloading**
- **Dynamic Polymorphism – Runtime**
 - Achieved via **Method Overriding**

3. Static Polymorphism (Method Overloading)

- Same method name, different parameter lists.
- Overloaded method is resolved at **compile time**.

Example:

```
class ManualCar {  
    void accelerate();           // no parameter  
    void accelerate(int speed); // with parameter  
};
```

- Allows the same behavior to adapt based on passed arguments.

Rules:

- Method name: Same
- Return type: Can be same or different (but not used for overloading)
- Parameters:
 - Vary in number or type

4. Dynamic Polymorphism (Method Overriding)

- Same method signature is redefined in child classes.
- Achieved using **virtual functions** in C++.
- Resolved at **runtime**.

Example:

```
class Car {  
    virtual void accelerate() = 0; // Abstract  
};  
  
class ManualCar : public Car {  
    void accelerate() override; // Manual-specific logic  
};  
  
class ElectricCar : public Car {  
    void accelerate() override; // Electric-specific logic  
};
```

5. Combined Use of OOP Pillars

- Final code demonstrates:

// See code section for full code example.

- **Abstraction** (Hiding implementation details)
- **Encapsulation** (Private/protected members)
- **Inheritance** (Manual/Electric inherit Car)
- **Polymorphism** (Method overriding & overloading)

Additional Concepts:

- **Protected:**
 - Inaccessible outside class, but accessible in child class.
- **Operator Overloading (Homework):**
 - Concept asked as homework: What is operator overloading?
 - Why is it available in C++ but not in Java/Python?

Conclusion & Practice

- Understanding OOPs is best done via **real-world relatable examples**.
- Practice suggestion: Modify/add features to existing car classes.

- **Homework:**

Operator overloading is a feature in C++ that allows developers to redefine the behavior of operators (like +, -, *, etc.). For example, you can overload the + operator to add two complex numbers or concatenate two strings.

1. **Define Operator Overloading.**
2. **Why is it not supported in Java/Python?**

In Java, the language designers chose to avoid operator overloading to prevent confusion and maintain a clear syntax. In Python, while it does support operator overloading, it is done through special methods (like `__add__`, `__sub__`, etc.) rather than allowing direct operator overloading as in C++.